

Traffic management system:

Real Time Traffic Management systems manage traffic behaviours in real time by utilising a network of technologies including sensors, smart cameras, global positioning systems (GPS) and Bluetooth/Wi-Fi. This can be used to efficiently reduce congestion, bottlenecks and other traffic issues. Real-time data can be used to suggest alternate routes to drivers when routes are congested and indicate to public transport operators and decision makers where user demand and supply is located. Technology improvements have allowed the development of sophisticated services to operate networks to resolve the conflicting demands of all road and transport users.

Code that create a platform that displays a real-time traffic information:

1.Set Up Flask Web App:

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    # Render the main page
    return render_template('index.html')

if __name__ == '__main__':
    app.run(debug=True)
```

2.Create HTML Template (templates/index.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Real-Time Traffic Information</title>
</head>
```

```
<body>
  <h1>Real-Time Traffic Information</h1>
  <div id="map"></div>
  <!-- Add any additional elements or scripts for the map -->
</body>
</html>
```

3.Integrate Map API:

We can use a mapping library like Leaflet or Mapbox to display real-time traffic information on a map. Include the necessary scripts in your HTML file.

4.Fetch Real-Time Traffic Data:

Use an API that provides real-time traffic data. For example, we can use the Google Maps API for this. we'll need an API key.

```
import requests

API_KEY = 'your_google_maps_api_key'
BASE_URL = 'https://maps.googleapis.com/maps/api/directions/json'

def get_traffic_data(origin, destination):
    params = {
        'origin': origin,
        'destination': destination,
        'key': API_KEY
    }

    response = requests.get(BASE_URL, params=params)
    data = response.json()

    # Extract relevant traffic information from the response
    # Process the data as needed
    return data
```

5.Update Frontend with Real-Time Data:

We Used JavaScript to periodically fetch and update the real-time traffic data on the frontend.

```
// Example using AJAX for asynchronous data fetching
function updateTrafficData() {
    // Make an AJAX request to your server to fetch real-time
    traffic data
    // Update the map or any other elements on the page
}

// Update every 5 minutes (adjust as needed)
setInterval(updateTrafficData, 300000);
```

6.Code for the IOS and android (React):

```
expo init TrafficApp
cd TrafficApp
npm install react-native-maps react-navigation
```

```
// MapScreen.js
import React from 'react';
import { View, StyleSheet } from 'react-native';
import MapView from 'react-native-maps';

export default function MapScreen() {
    return (
        <View style={styles.container}>
            <MapView
                style={styles.map}
                initialRegion={{
                    latitude: 37.7749,
                    longitude: -122.4194,
                    latitudeDelta: 0.0922,
                    longitudeDelta: 0.0421,
                }}
            />
        </View>
    );
}
```

```

const styles = StyleSheet.create({
  container: {
    flex: 1,
  },
  map: {
    flex: 1,
  },
});
// AppNavigator.js
import { createAppContainer } from 'react-navigation';
import { createStackNavigator } from 'react-navigation-stack';
import MapScreen from '../screens/MapScreen';

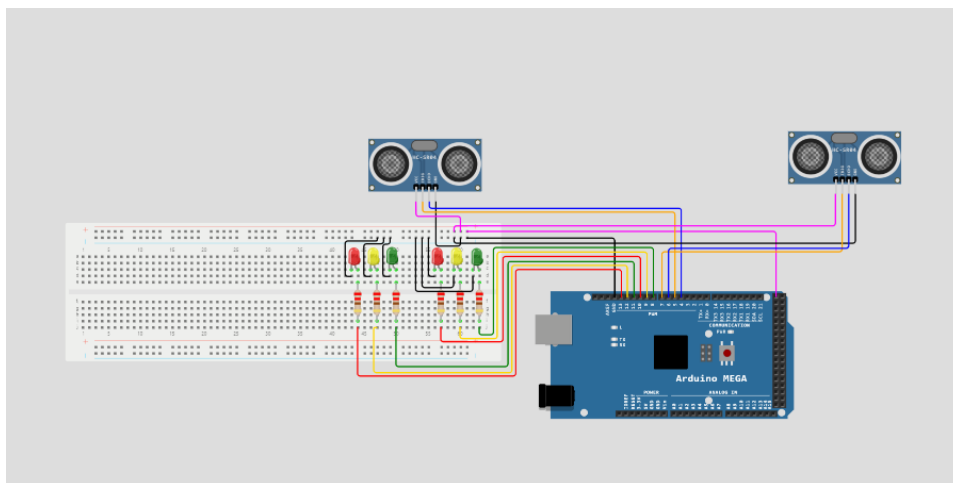
const AppNavigator = createStackNavigator({
  Map: MapScreen,
});

export default createAppContainer(AppNavigator);
// App.js
import React from 'react';
import AppNavigator from './navigation/AppNavigator';

export default function App() {
  return <AppNavigator />;
}

```

Structured image:



The IoT Traffic Monitoring System offers a technological solution to the challenges of urban traffic, providing decision-makers with valuable insights for better traffic management and fostering a data-driven approach to city planning. The project represents a step towards smart cities that leverage IoT technologies to create more efficient and sustainable urban environments.